Muhammad Usman Aziz

(+92)3216692660 usmanamjad495@gmail.com LinkedIn GitHub

OBJECTIVE

AI and Full-Stack Developer specializing in Generative AI, Deep Learning, and Web Technologies. Focused on developing AI-powered applications, ERP automation, improving accessibility, and building scalable real-time systems. Eager to contribute expertise to innovative projects at the intersection of AI, enterprise ERP systems, and modern web development.

EDUCATION

Bachelor of Computer Science, FAST-NUCES, CFD Campus — Sep 2021 - June 2025

SKILLS

Programming Languages	Python, TypeScript, JavaScript, C++, SQL, R, HTML5/CSS3
AI & Machine Learning	Generative AI, LLMs, RAG, Fine-Tuning (LoRA/PEFT), NLP, Deep Learning, PyTorch, TensorFlow, OpenCV, Keras, Transformers, Pandas, NumPy, Scikit-learn
Full-Stack & Web	React.js, TanStack Start, Node.js, Express.js, Frappe / ERPNext, Tailwind CSS, REST APIs, Webhooks, Dexie (IndexedDB), Cloudflare Workers
Mobile Development	React Native, Capacitor, Android Studio, Expo Go
Databases & ERP	MariaDB, MySQL, MongoDB, ERPNext, Google Sheets API, QuickBooks Integration
Developer Tools	Git, GitHub, VS Code, Cursor, Postman, Vite, Jupyter Notebook, Colab, PyCharm

EXPERIENCE

Software Engineer — Oct 2025 - Present
Agrovisions — *Faisalabad, Punjab, Pakistan*

- Developed and customized enterprise ERPNext solutions, automating sales workflows, live stock tracking, and financial invoicing.
- Engineered tamper-proof cryptographic document verification systems and dynamic QR code generation for authentic corporate records.
- Spearheaded automated ETL pipelines to extract, transform, and migrate multi-year legacy QuickBooks Desktop financial data into ERPNext.
- Built and maintained full-stack web applications and REST API integrations connecting external e-commerce platforms with ERPNext.

Computer Science & STEM Instructor — Aug 2019 - Present
The Century Academy of Science Commerce and Arts — *Faisalabad, Punjab, Pakistan*

- 7+ years of experience teaching Computer Science, Mathematics, and Physics to high school and intermediate students.
- Mentored students in core programming fundamentals, algorithmic problem solving, and analytical thinking with excellent academic outcomes.

Artificial Intelligence Intern — July 2024 - Aug 2024
CodeAlpha — *Remote*

- Developed AI-powered Autonomous Robot Delivering in Python.

Data Analyst Intern — July 2024 - Aug 2024
Oasis Infobyte — *Remote*

- Exploratory Data Analysis (EDA) on Retail Sales Data. Used K-means clustering for segmentation.

Machine Learning Intern — Aug 2024 - Sep 2024
GrowIntern — *Remote*

- Applying ML techniques to real-world problems and gaining hands-on experience with model development and deployment.

PROJECTS

M Medium Blogs

Stockly - (TanStack Start, React, TypeScript, Tailwind CSS, Dexie, Capacitor)

- Developed an offline-first inventory, sales, and profit management application with IndexedDB persistence, automated Google Sheets OAuth sync, and cross-platform mobile support via Capacitor.

QuickBooks Desktop Data Extractor - (Python, QBXML SDK, VBScript COM, Web UI)

- Developed a production-grade extraction engine leveraging QBXML and COM Bridge to export complete company master data (COA, Customers, Vendors) and 20+ transaction types into migration-ready JSON.

QuickBooks to ERPNext Migration Engine - (Frappe, ERPNext, Python, Pandas)

- Engineered an automated ETL migration pipeline and custom Frappe application to parse, map, and import historical QuickBooks accounting, payroll, and inventory records into ERPNext DocTypes.

ERPNext Notification & Workflow Engine - (Frappe, ERPNext, Jinja2, Python)

- Designed dynamic notification templates and automated task/todo triggers in ERPNext to optimize enterprise-wide inter-department communication and assignment workflows.

ERPNext Document Verification Portal - (ERPNext, Frappe, Python, JavaScript, QR Code)

- Built an enterprise document authentication system for ERPNext invoices and certificates featuring tamper-proof cryptographic QR codes and a dedicated public verification portal.

Result Card Generator - (TanStack Start, React, TypeScript, Tailwind CSS)

- Developed an automated school result card generator featuring pixel-perfect A4 printing layouts, automatic GPA and percentage calculations, and a fully responsive preview interface.

Oxigen E-Commerce Platform - (React, Node.js, Express, ERPNext REST API, Tailwind CSS)

- Created a high-performance retail e-commerce platform and admin portal integrated with ERPNext for real-time inventory management, automated order routing, and invoice synchronization.

Shopify AI Assistant - (Node.js, Express, LLM API, Shopify Storefront API)

- Built an intelligent AI chatbot for Shopify storefronts handling automated 24/7 customer support, semantic product search, and live order status inquiries.

Gemini Web 2 API - (TypeScript, Node.js, REST API)

- Engineered an API gateway exposing web-based Gemini AI capabilities as OpenAI-compatible REST endpoints with session management and multi-turn conversational support.

Deaf-ED (FYP) - (MERN, OpenCV, and LSTM)

- Designed a mobile app using PSL and animations to teach Urdu, English, and basic counting, featuring real-time sign assessment and parental progress tracking.

Shayar AI - (LSTM, GRU, RNN)

- Shayar AI is an intelligent Urdu poetry generator that crafts original verses from a single input word. It allows users to interactively discover fresh poetic expressions, offering a unique and immersive literary experience.

Medical AI Assistant - (RAG)

- Smart medical assistant that mimics doctor-patient chats, retrieves cases, and creates detailed medical reports.

Fine-tuning DeepSeek-Math - (Fine-Tuning, LoRA)

- Solve emoji-based math problems using Parameter-Efficient Fine-Tuning (PEFT) with LoRA.

Pseudocode → C++ / C++ → Pseudocode - (Seq2Seq model, Transformers)

- Trained Transformers from scratch to convert pseudocode to C++ code and vice versa for educational use.

Autonomous Robot Delivering – (A*, BFS, Pygame)

- Simulated a 15×15 grid-based delivery system with dynamic obstacles and randomized traversal costs using A* and BFS algorithms.

Royal Decore – (HTML, CSS, JavaScript)

- Built a responsive home décor e-commerce website with product listings, interactive UI, and basic cart functionality using vanilla web technologies.

Event Horizon – (Node.js, Express.js, MongoDB)

- Developed a full-stack event management platform with dynamic event listings, user registration, and backend API integration using Node.js, Express, and MongoDB.

AI-Powered Math Riddle Generator – (Python, Streamlit, GPT-2)

- Built a Streamlit web app that generates unique math riddles using a fine-tuned GPT-2 model trained on a dataset of mathematical puzzles based on custom user prompts.

AI-Powered Rap Poetry Generator – (Python, Whisper, Gemini API, Play.ht)

- Built an AI app that transcribes voice input using Whisper, generates rhyming poetry using Gemini API, and converts it to speech with Play.ht.

Math Meme Repair – (Python, DeepSeek Math 7B, LoRA)

- Fine-tuned DeepSeek Math 7B using LoRA to correct errors in viral math memes and rate responses with a playful “sass vs. patience” scale.